Digital Avionics
航電系統

Lecture 1

Prof. Ying-Chih Lai

Outline

- Course Information
- Course Syllabus(課程大綱)
- Course Objectives(課程目標)
- Introduction to Avionics Systems

Course Information - People

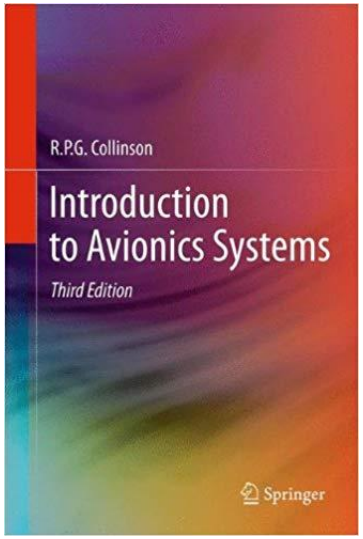
- Instructor: Lai, Ying-Chih (賴盈誌)
  - Office: 5854
  - Email: yingclai@mail.ncku.edu.tw
  - Office hour: 2:00 - 4:00 PM on Thursday

Course Information - Prerequisites

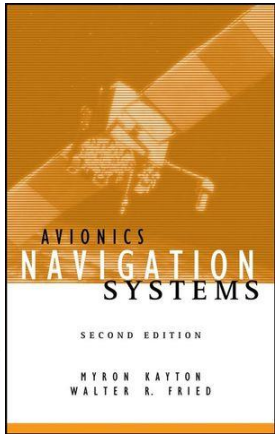
- Preparatory Course
  - Introduction Civil Aeronautics
  - Introduction to Aeronautical Engineering
  - Aircraft Design
- Desirable (but not necessary)
  - Classic Control Theory
  - Flight Mechanics

Course Information - Textbook

- Collinson, Richard PG. *Introduction to avionics systems*. Springer Science & Business Media, 2013. (Third Edition)
- Download Link
- References:
  - 林清一，數位航空電子系統(Digital Avionics Systems)，全華圖書，2017
  - Kayton, Myron, and Walter R. Fried. *Avionics navigation systems*. John Wiley & Sons, 1997. ISBN: 0471547956
  - Nelson, Robert C. *Flight stability and automatic control*. Vol. 2. WCB/McGraw Hill, 1998.







Course Information - Textbook

- *Introduction to Avionics Systems (Third Edition)*
  - Chapter 1 - Introduction
  - Chapter 2 - Displays and Man–Machine Interaction
  - Chapter 3 - Aerodynamics and Aircraft Control
  - Chapter 4 - Fly-by-Wire Flight Control
  - Chapter 5 - Inertial Sensors and Attitude Derivation
  - Chapter 6 - Navigation Systems
  - Chapter 7 - Air Data and Air Data Systems
  - Chapter 8 - Autopilots and Flight Management Systems
  - Chapter 9 - Avionics Systems Integration
  - Chapter 10 - Unmanned Air Vehicles

Course Syllabus

- Tentative Syllabus

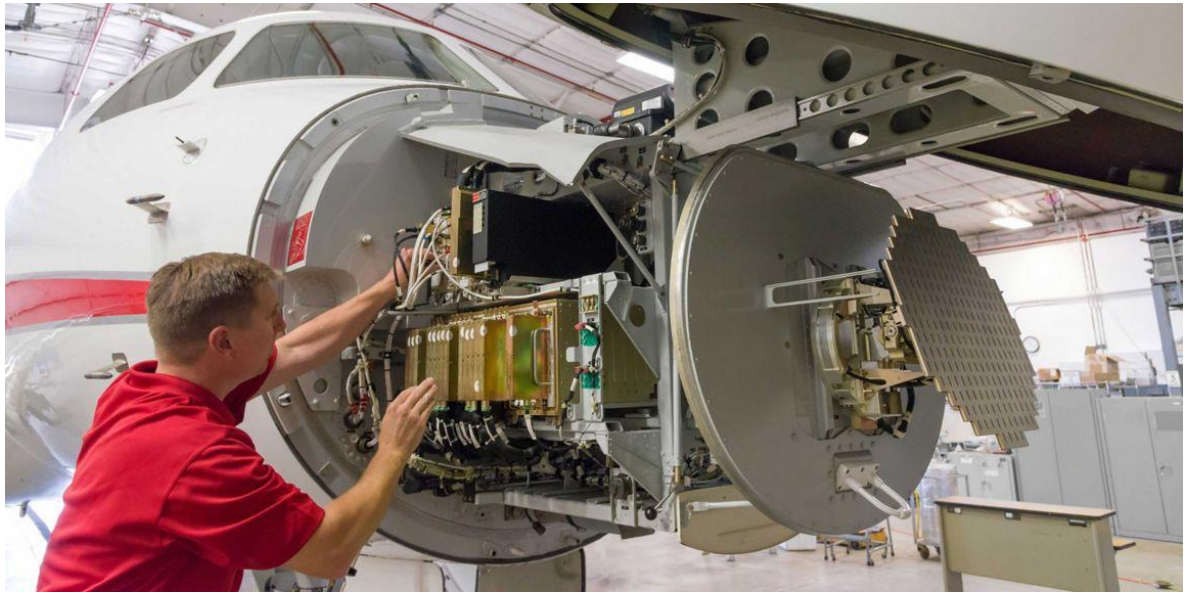
Week	Title of Unit and Content	Assignments/Quizzes
1	Introduction to Digital Avionics	Classes begin
2	Aerodynamics and flight control	
3	Aerodynamics and flight control	Assignment 1
4	Flight Control System	
5	Flight By Wire System	Assignment 2
6	Inertia Navigation System	
7	Inertia Navigation System	Assignment 3
8	Global Positioning System	
9	Midterm Exam (Nov. 4)	
10	Global Positioning System	
11	Air Data System	Assignment 4
12	Air Data System	
13	Airborne Radar & Ground Proximity Warning System	Assignment 5
14	Traffic Alert and Collision Avoidance System	
15	Autopilot and Flight Management	Assignment 6
16	Flight Simulator Practice	
17	Term Project Presentation (Dec. 30)	
18	Final Exam (Jan 6)	

Course Information - Policies

- Midterm Exam(期中考) 30%
- Term Project(期末專題報告) 15%
  - 2-3 people in a team
- Final Exam(期末考) 30%
- Assignment(作業) 15%
  - 4-6 assignments
  - Submit one week later after assigning
  - Late assignments only get part scores
  - Don’t copy! If you do this, you will get nothing.
- Class Participation(平時成績) 10%
  - Class attendance and attitude
  - Turn off the smartphones or change to vibration mode

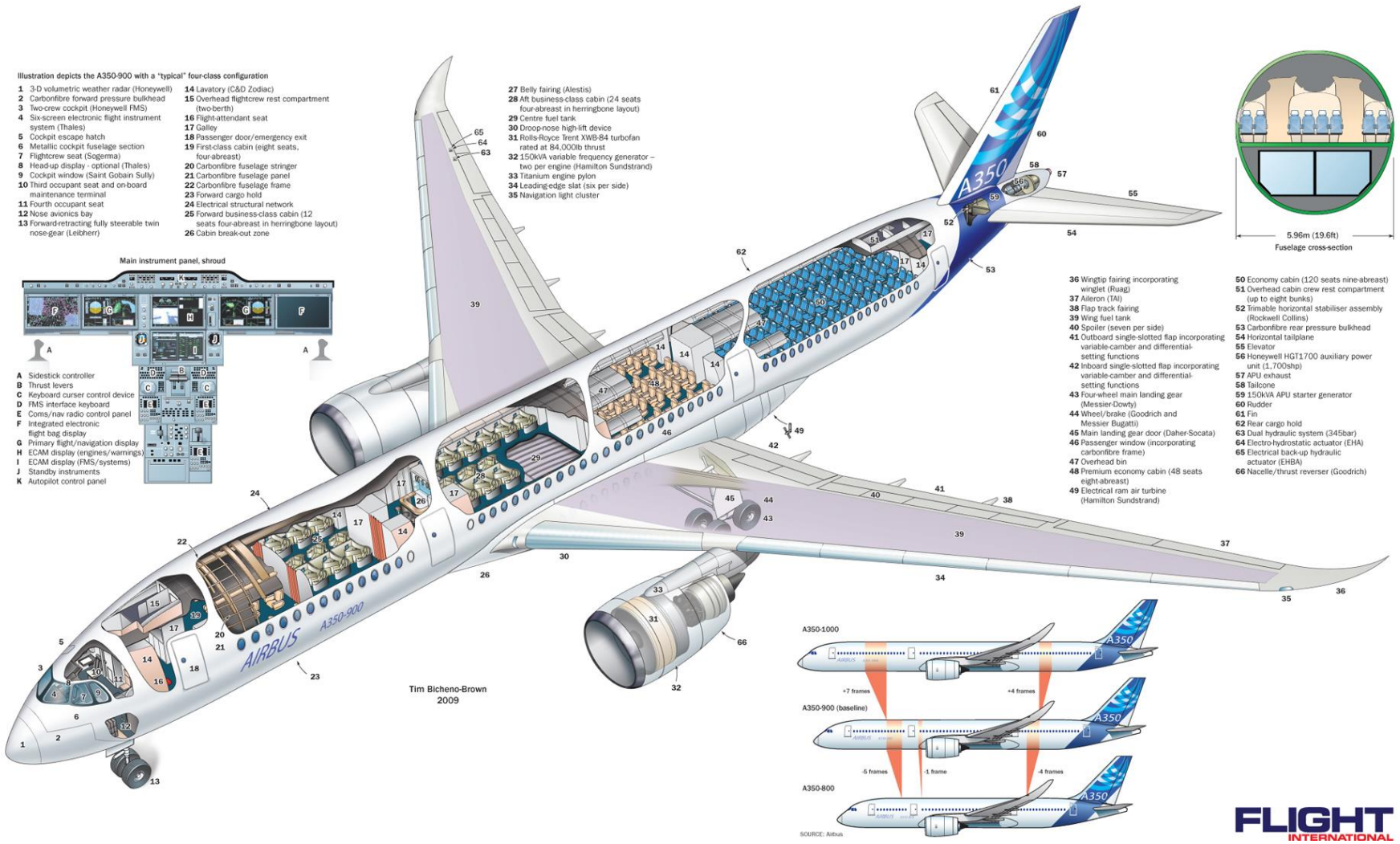
Course Objectives

- Establish fundamental concept on avionics systems, offer the principles, design considerations, and flight operation of flight instruments, together with data processing for system integration.



https://aerospace.honeywell.com/us/en/about-us/blogs/can-you-afford-to-self-insure-business-aircraft-maintenance

What are avionics systems ?



What are avionics systems ?

- Avionics are the electronic systems used on aircraft, artificial satellites, and spacecraft. Avionic systems include communications, navigation, the display and management of multiple systems, and the hundreds of systems that are fitted to aircraft to perform individual functions.

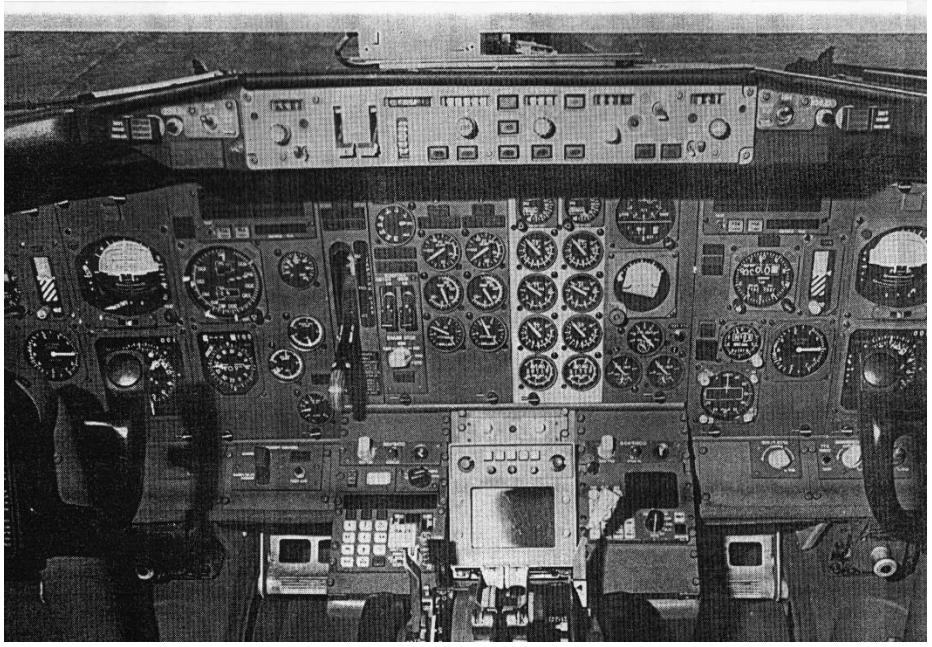




F-105 Thunderchief with avionics laid out

https://en.wikipedia.org/wiki/Avionics

What are avionics systems ?



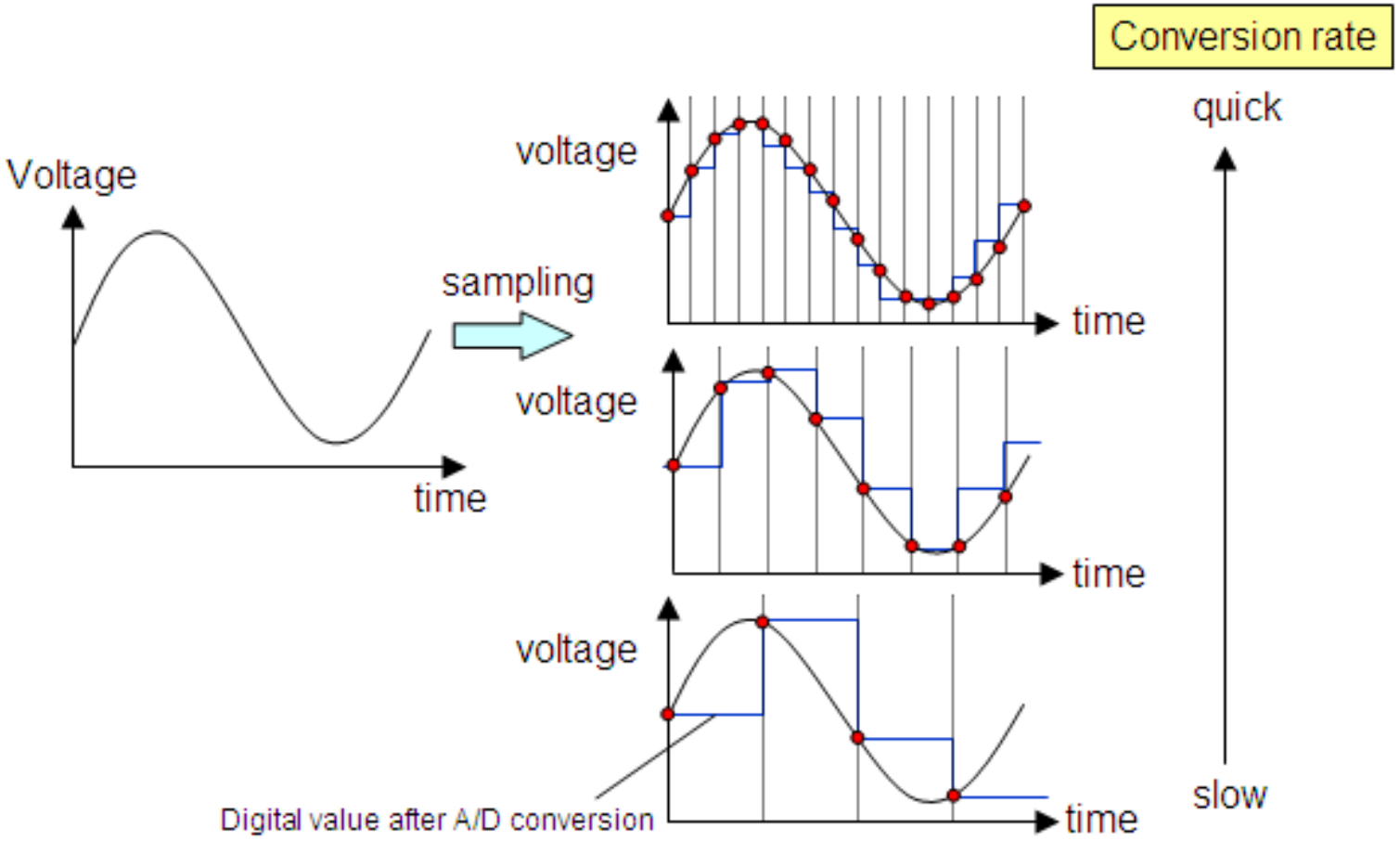
B-737 Analog Avionics



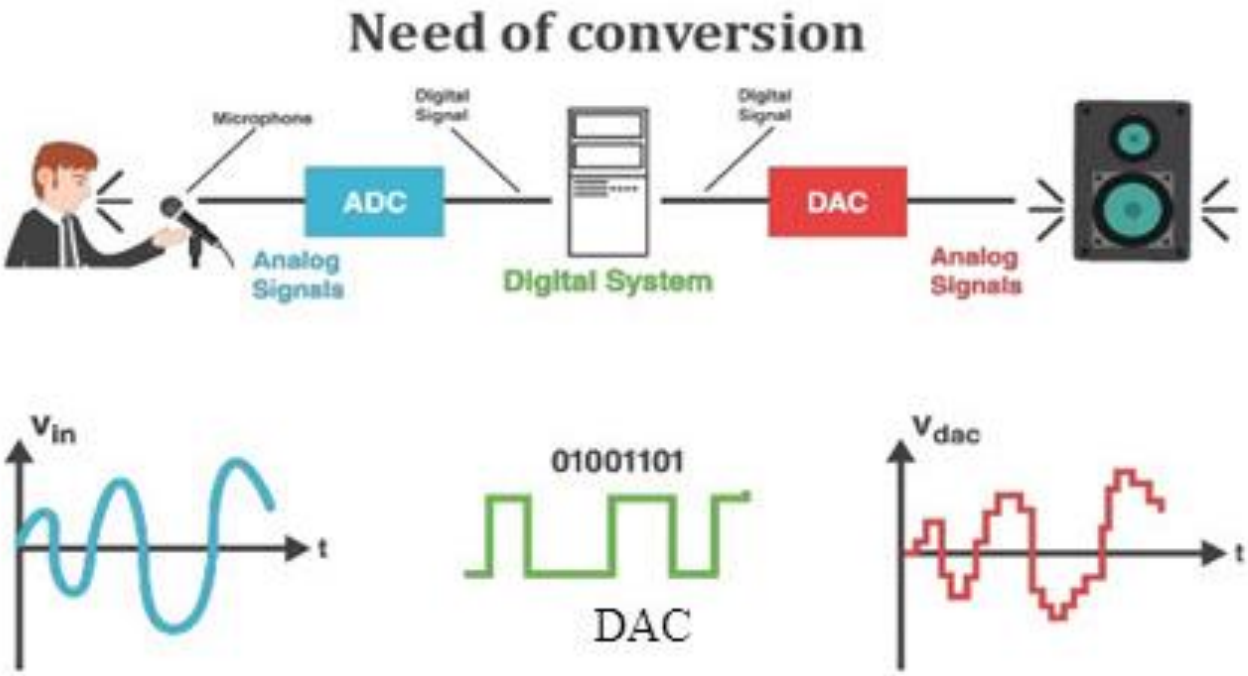
B-737 max Digital Avionics


BEFORE
IN PROGRESS
AFTER

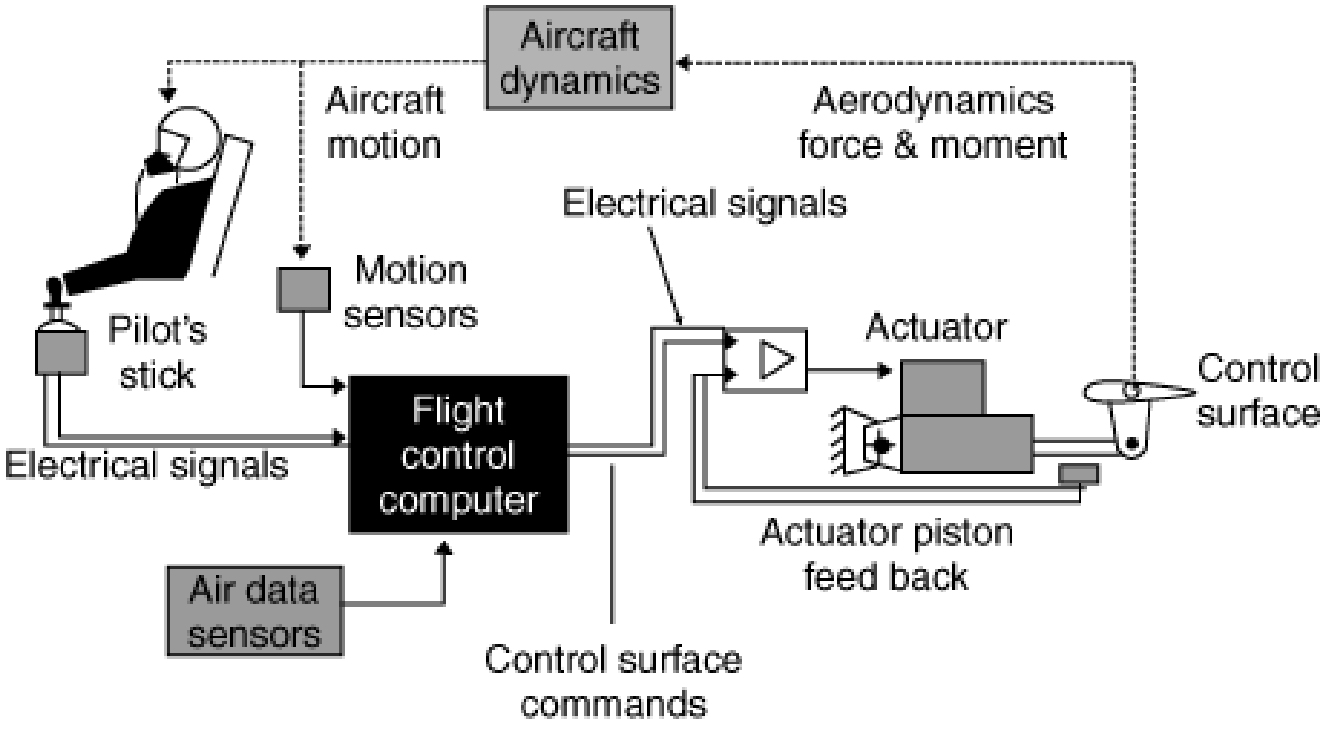
Analog to Digital Converter



Digital to Analog Converter

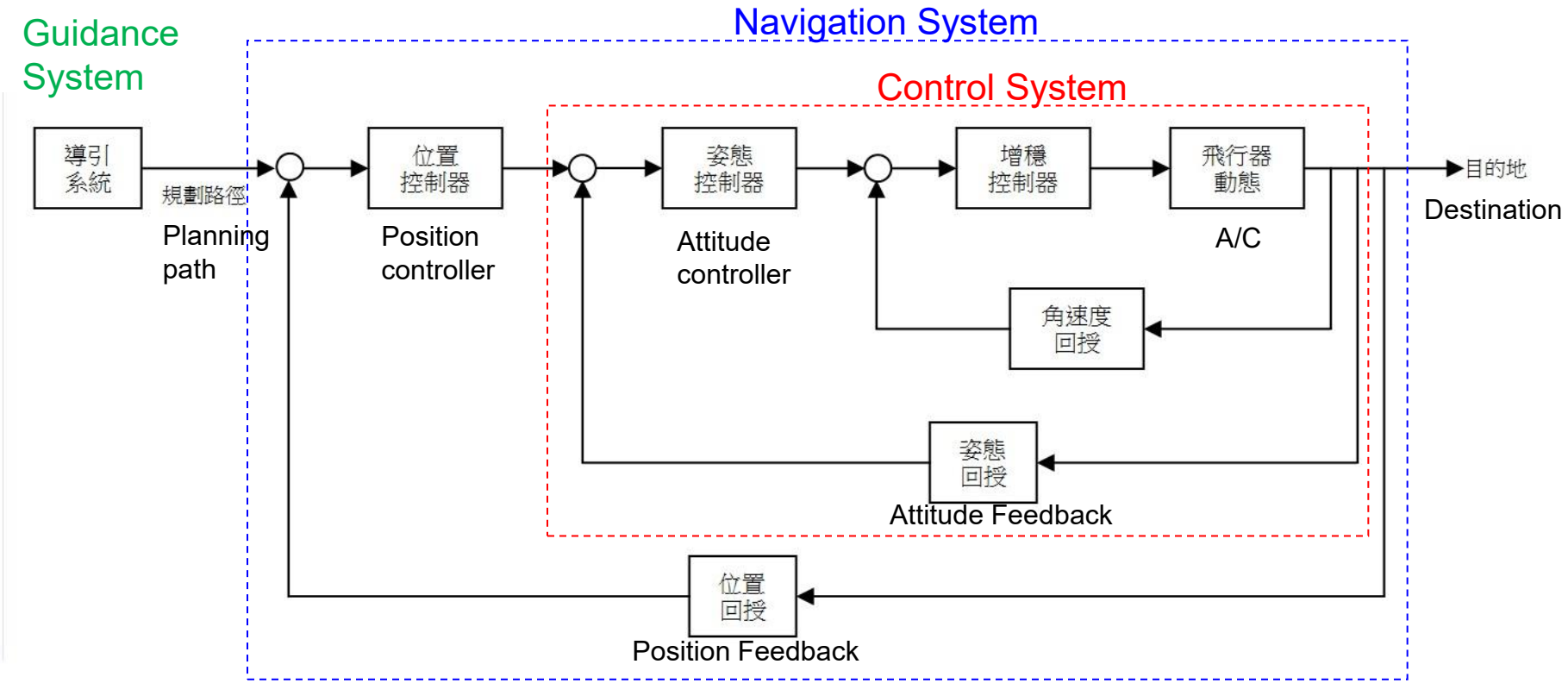


What are avionics systems ?



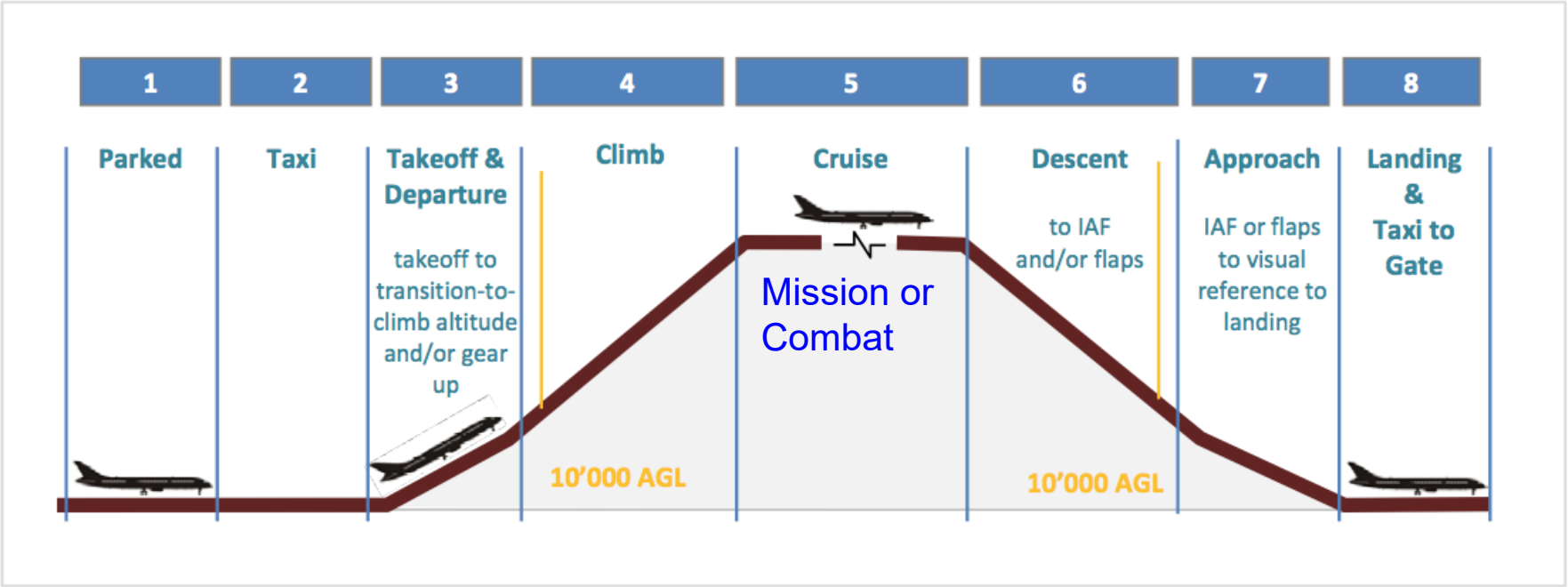
Guidance, Navigation, and Control of an Aircraft

- System Block Diagram



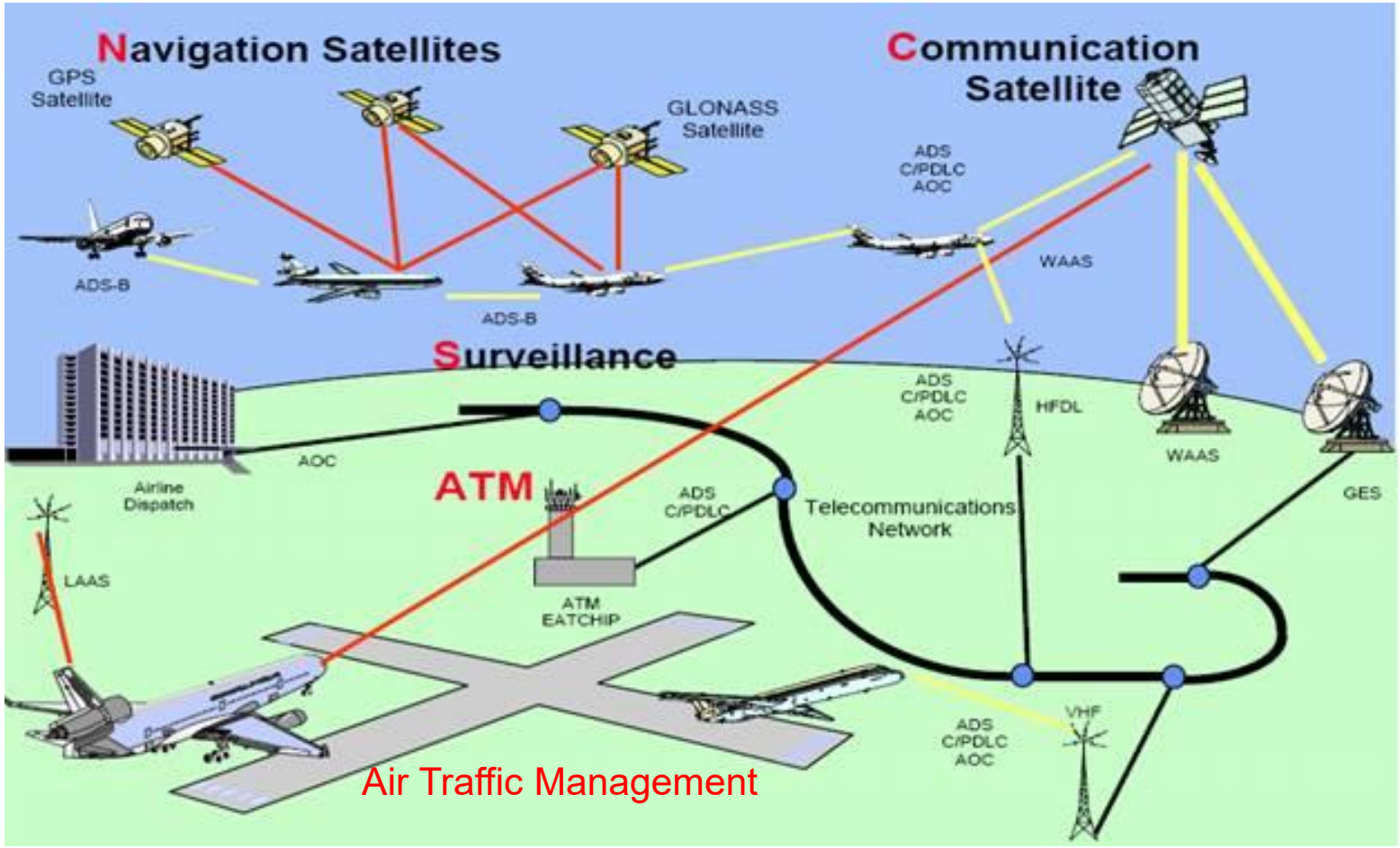
Phases of Flight

- Flight Simulator Training (Homework)

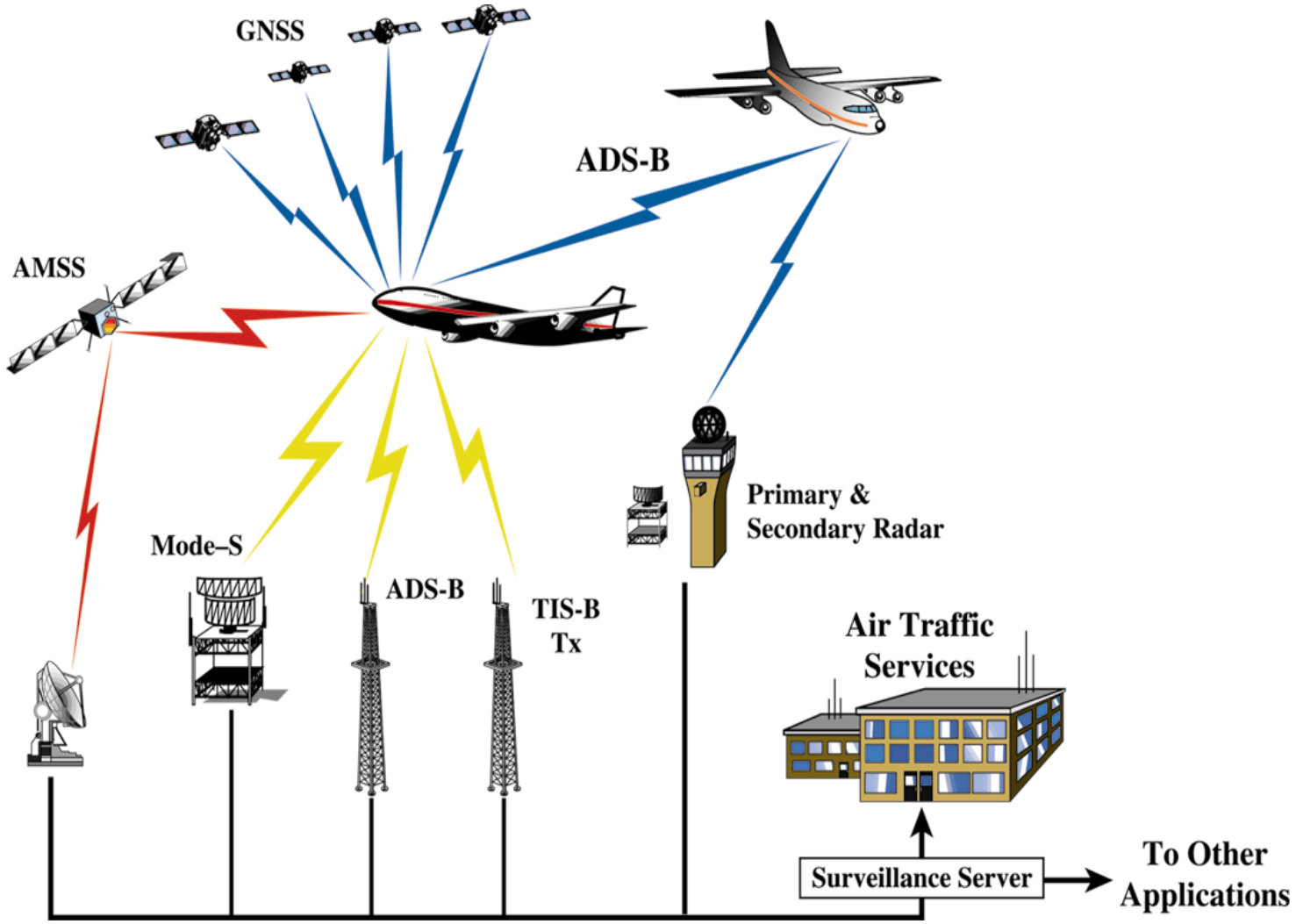


How to reach the destination ?

- **CNS/ATM**

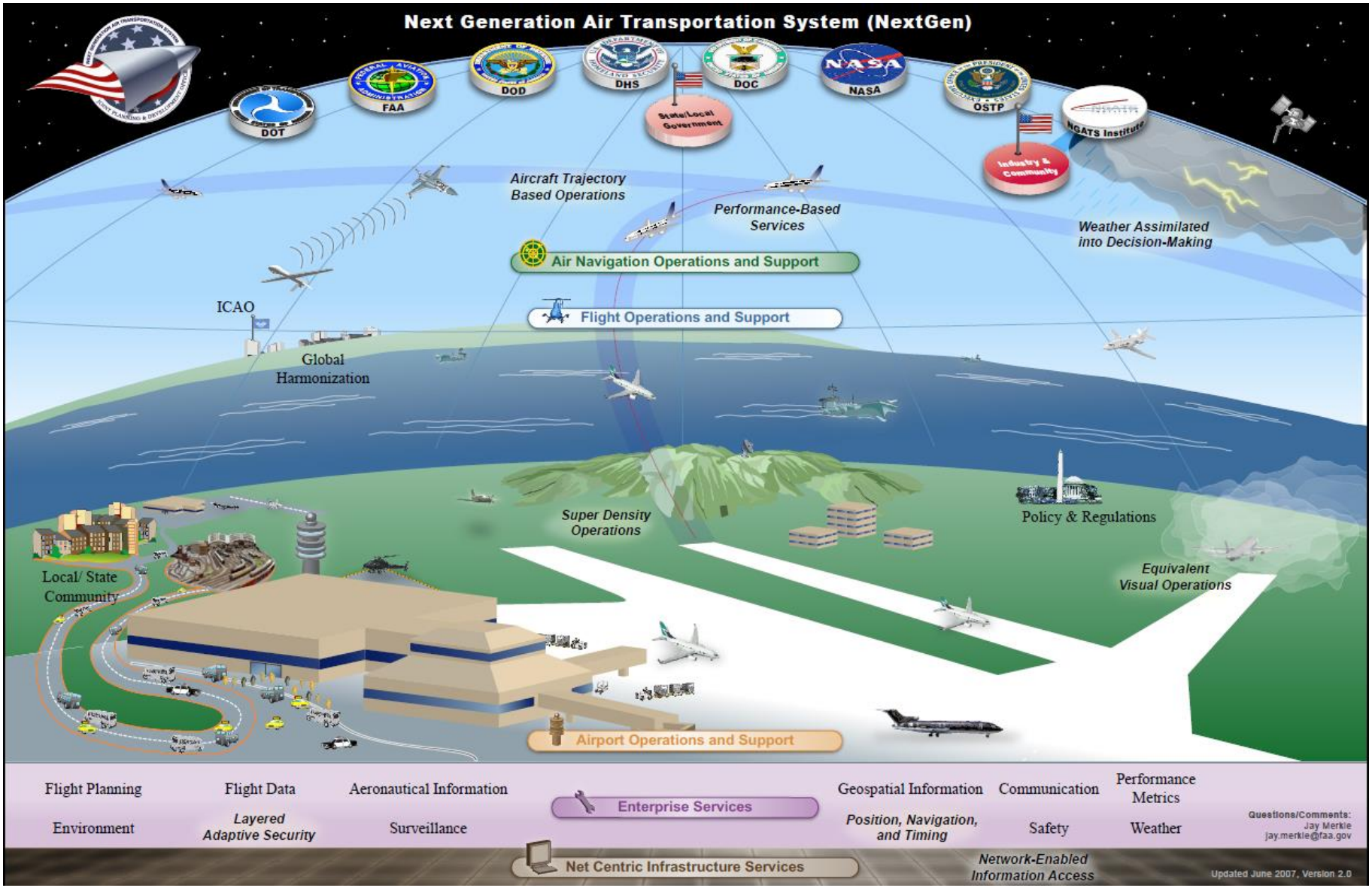


Broad Navigation Avionics



FAA's NextGen Project

https://www.youtube.com/watch?v=fadkI4JJfoE



FAA's NextGen Project

https://www.faa.gov/nextgen/reporting-benefits

Between 2010 and 2025 implemented capabilities generated **$13.9 Billion** **in benefits** *(in 2025 dollars).*

Safety Benefits

encompass



$0.7 Billion



Fuel Savings

amount to



$2.5 Billion



Aircraft Operating Cost Savings

come to



$2.7 Billion



Passenger Travel Time Savings*

make up



$8 Billion

